CPSC 314 Final Project

PROJECT PLAN
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Sumo Robots

Team Sumo

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1 Project Overview

1.1 Project Summary

Our team is developing an autonomous sumo combat system designed to demonstrate tactical navigation and mechanical leverage within a competitive arena. The primary goal of this mission is to create a robot that can independently locate an opponent and push them out of a circular ring. To gain a physical advantage, we have designed custom 3D-printed ramp attachments for the front of our chassis. These ramps are intended to slightly lift the opposing robots' wheels to reduce their friction against the floor, making it much easier for the attacking robot to push the other out of the ring.

The project involves two identical autonomous robots, each controlled by a Raspberry Pi 5 paired with a Yahboom Raspbot v2 expansion board. To navigate the arena, each robot relies on two primary sensing methods. An ultrasonic sensor is mounted on the front to act as the robot's "eyes," scanning the ring to find the opponent through sound wave reflection. Simultaneously, a 4-channel infrared sensor is pointed at the floor to monitor the arena's surface. This ground sensor is specifically programmed to detect the black tape that marks the edge of the ring. In our programming, if this sensor reads the black tape, it signifies that the robot has reached the boundary and has lost the match, while the robot that remains inside the circle is declared the winner.

2 Project Requirements

2.1 Major Features

Table 1: Major Features

Feature	*Description*
Autonomous Driving	The autonomous driving feature allows the motors to operate entirely on their own based on pre-programmed logic rather than manual remote control. Its technical significance lies in the robot's ability to make real-time navigation decisions without human intervention, which is essential for the fast-paced nature of sumo combat. This ensures the system can immediately switch between searching for an opponent and engaging in a push.
Ultrasonic Detection	This feature uses high-frequency sound waves to measure the distance between our robot and the opponent. Technically, it serves as the primary "vision" for the system, allowing the Raspberry Pi 5 to calculate exactly where the other robot is located within the ring. Without this sensor, the robot would be unable to actively target the opponent and would have to rely on random movement.
Black Tape Detection	Black tape detection uses downward-facing infrared sensors to monitor the floor for the arena's boundary lines. Its technical significance is acting as a fail-safe that prevents the robot from driving itself out of the ring and losing the match by mistake. These sensors provide the critical data needed to trigger the "loss" state in the software logic.

3D printed hardware	Our custom 3D-printed ramps are physical attachments designed to give the robot a mechanical advantage during a collision. Technically, these ramps are significant because they alter the physics of the fight by getting underneath the opponent's chassis to break their traction. This allows our robot to utilize its motor torque more effectively to win the match. They also provide protection to the 4-channel tracking sensor (used for finding black tape) that sticks out in front of the robots' chassis.
Win / Loss communication	This feature uses integrated LEDs to provide an immediate visual output based on the match's outcome. The technical significance is the implementation of a conditional logic loop: if the black tape is detected, the losing robot's LEDs turn red, while the other robot triggers a green LED signal. This translates internal sensor data into a clear, visible result for the users and spectators.

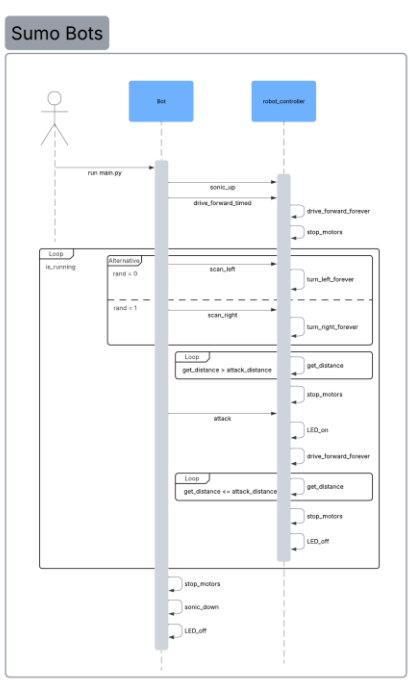
3 Project System Design

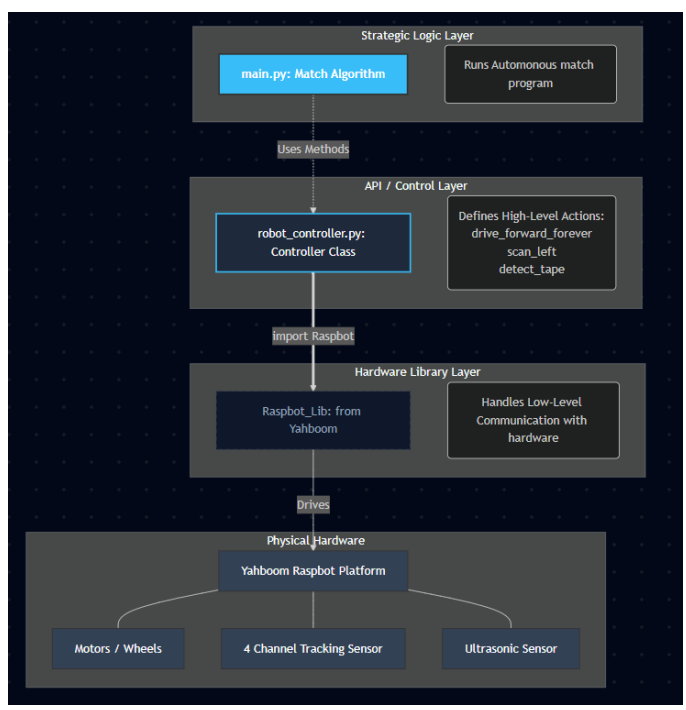
3.1 Initial User Interface Design

Since both of our robots will be fully autonomous, we will not be developing a control console for either of our robots.

3.2 Initial Software Architecture

We will build our software on the raspbot_lib API, a set of low-level methods to control the hardware of the raspbots. For simplicity, we will create a wrapper API called robot_controller that implements high-level robot control functions that call methods from the raspbot_lib API. The main logic for sumo bots will be contained in main.py. In main.py, we will instantiate a robot object capable of calling methods from the robot_controller API. In this way, our main logic will be simple, high-level method calls that perform complex, low-level raspbot_lib functions behind the scenes, enabling us to develop sound logic without the distraction of dealing with complex, low-level raspbot_lib calls.





4 Project Schedule

Table 3: Major Scheduling Milestones

Milestone	*Description*	*Target Completion Date*
Assemble Robots	Both robots are fully assembled and ready to begin programming / measuring for the 3d printed ramps	*March 1*
Set up coding environment	Both robots have the required software and tools and are fully set up to begin coding	*March 8*
Robot movement demo	Robots are able to perform basic autonomous movement	*March 15*
Robot_controller API set up	Robot_controller is set up to begin coding robot methods within it. Code from the movement demo should be migrated over to ensure functionality	*March 22*
Ultrasonic sensor methods	Methods to control and receive data from the ultrasonic sensor are functional in robot_controller	*March 29*
IR sensor methods	Methods to control and receive data from the front IR sensors are functional robot_controller	*April 5*
LED methods	Methods to control the rear LEDs are functional in robot_controller	*April 12*
3D printed ramps	First rendition of 3D printed ramps are complete. Now that IR sensors are protected, robots can be tested against each other to create the sumo logic	*April 19*
Main.py logic	Sumo logic is complete; robots successfully compete in sumo battle against one another	*April 26*
Code complete	Code is polished and bug-free	*April 27*
Presentation	Penetration day	*May 1*
Project plan	Project plan document complete	*May 8*
Final report	Final report document complete	*May 8*